

Kayleen Cruz

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EDUCATION

Brown University, Providence, RI

Expected Graduation Date: May 2026

Bachelor of Science in Computer Science, GPA: 3.71

Relevant Courses: Data Structures and Algorithms, Computer Systems, Software Engineering (in progress), Embedded Software (in progress), Discrete Structures

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, HTML, CSS, Javascript, Typescript, ReactJS

Developer Tools/Frameworks: Visual Studio Code, IntelliJ, Github, Git, Firebase, Bootstrap, TailwindCSS, Docker

PROJECTS

CSV Parser and Census Data Retrieval API (Java, ReactJS, Typescript, Clerk, Firebase)

- Developed an extensible CSV parser library, increasing flexibility and broad usability across applications.
- Built a RESTful API server supporting data retrieval and search functionality, with efficient caching to integrate real-time Census data for broadband accessibility insights.
- Designed a web application frontend that connects with the API providing users an interactive interface to load, view, and search CSV data, with backend integration for broadband data visualization.

Caching I/O (C, Docker)

- I/O library that is ~21,000x faster than direct system calls and within 3x the stdio library function calls by implementing a cache to reduce the number of disk operations required.

Distributed Store (C, Docker)

- Built a scalable, sharded key-value storage system with a centralized shard controller to distribute data across multiple servers.
- Developed a sharding-aware client and server to ensure efficient data access and distribution.

WeensyOS (C, Docker)

- Implemented virtual memory architecture with memory isolation, process creation, and cleanup for a small OS

WORK EXPERIENCE

Cyborg Mobile and Microsoft | *New Technologist Intern*

June 2024 - August 2024

- Immersed in a team of 5 people to ideate, design, and engineer a full-stack web application.
- Collaborated with SWEs and PMs to complete the product development life cycle including intensive user and market research, user empathy, spec development, and prototyping.
- Led the frontend development by designing and implementing the application's user interface and layout.

Brown University Department of Computer Science | *Teaching Assistant*

January 2024 - May 2024

- Led debugging and conceptual hours where I provided individual support and guidance to students.
- Managed a lab section where I reviewed the key concepts discussed in lecture that week and aided exploration.
- Analyzed and evaluated students' design choices, functionality, and tests when grading and giving feedback.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Management Leadership for Tomorrow | *Career Preparation Fellow*

January 2024 - Present

Ignite CS | *Raices Dual Language Academy - Kids Who Code Leader*

September 2023 - Present

- Teach computer science through the programming language Scratch to local elementary school students.
- Engage in conversations regarding curriculum development and methods to effectively simplify abstract concepts.

Orientation Leader | *New Brunonians Orientation Welcoming Committee*

August 2023 - September 2023

ADDITIONAL SKILLS AND INTERESTS

Spanish, Product Management, Interested in machine learning, full-stack web development, and software engineering